

INDIAN INSTITUTE OF TECHNOLOGY DELHI
INTRODUCTION TO ELECTRICAL ENGINEERING
| ELL101 QUIZ-1 |

NAME:

DATE: 06-02-2025

Total marks: 15

Each question carries equal marks.

ENTRY NO:

DURATION: 45 min

SET: A

1. The current flowing through 8Ω resistor in Fig. 1 will be

A. 0.5 A B. 0.25 A C. 1 A D. 1.5 A

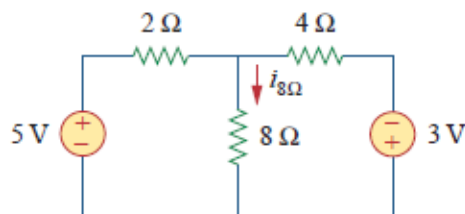


Fig 1

2. The value of R_{eq} for the given Fig. 2 will be

A. 6Ω B. 5Ω C. 8Ω D. 4Ω

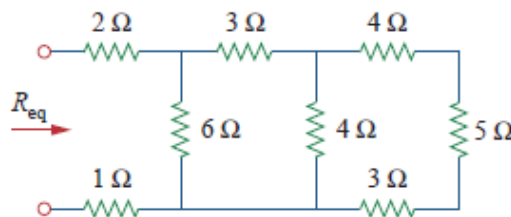


Fig 2

3. The voltage at node 1 in Fig. 3 as shown below will be

A. 40 V B. -20 V C. -40 V D. 30 V

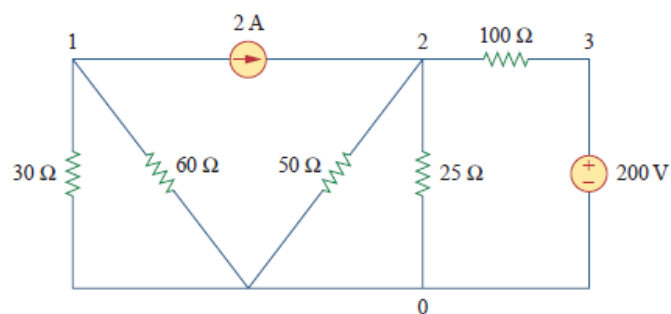


Fig. 3

4. The value of Norton Resistance (R_N) and Norton current (I_N) across terminal a-b will be (Refer Fig. 4)

A. $R_N = 7\Omega$, $I_N = 5\text{ A}$
 C. $R_N = 5\Omega$, $I_N = 7\text{ A}$

B. $R_N = 6\Omega$, $I_N = 8\text{ A}$
 D. $R_N = 8\Omega$, $I_N = 6\text{ A}$

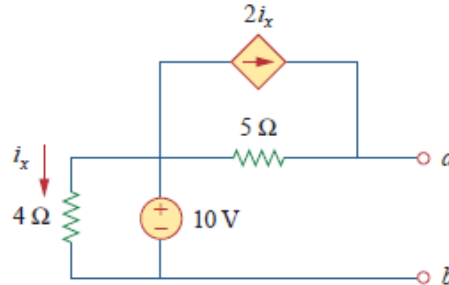


Fig. 4

5. For what Value of R , maximum power transferred to it from the rest of the circuit as in Fig. 5

A. 20Ω

B. 15Ω

C. 25Ω

D. 10Ω

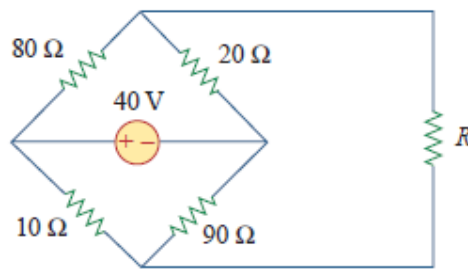


Fig. 5

6. The current flowing through a linear and active element is $i = \cos 4t$ and voltage across that element is $v = \sin 4t$ then the element would be

A. Resistor
 B. Capacitor
 C. Inductor
 D. Can be any one of the above 3

7. The circuit shown below in Fig. 6 is said to be in balanced condition if there is no current flow in the load resistance R_L . This will happen when

- A. $R_1 R_3 = R_2 R_4$
- B. $R_1 + R_2 = R_3 + R_4$
- C. $R_1 R_4 = R_2 R_3$
- D. $R_1 + R_4 = R_2 + R_3$

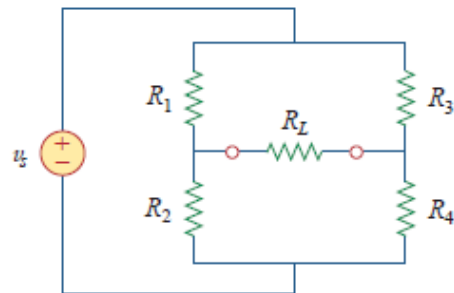


Fig. 6

8. A resistor develops 300 J of thermal energy in 15 second when 2 A current flows through it. If the current increases to 3 A, the energy developed (in Joule) in 10 second will be

- A. 350
- B. 250
- C. 500
- D. 450

9. A 220 V, 100 W bulb is connected to a 110 V source. Calculate the power consumed by the bulb

- A. 10 W
- B. 15 W
- C. 20 W
- D. 25 W

10. In the circuit below (Fig. 7), $R = 0$ and i_1 and i_2 are unknown. Find i .

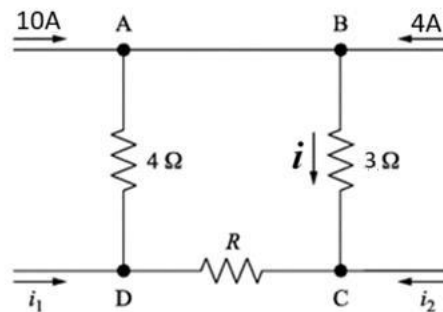


Fig. 7

- A. -8 A
- B. 24/7 A
- C. 8 A
- D. -24/7 A

11. In the given circuit (Fig. 8), superposition is applied. When V_2 is set to 0V, the current is i_2 is -6 A. When V_1 is set to 0V, the current i_1 is $+6$ A. What will the current i_3 when both sources will be applied?

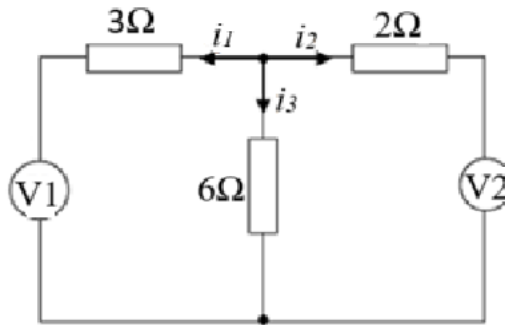


Fig. 8

- A. 1 A B. 5 A C. 3A D. 2A
12. Kirchoff's Current Law (KCL) states that (select one)
- A. The sum of voltage entering a junction equals the sum of voltages leaving the junction.
 - B. The sum of currents entering a junction equals the sum of current leaving the junction.
 - C. The voltage across a resistor is proportional to the current flowing through it
 - D. The potential difference across a capacitor remains constant over time.

